

Invoice and receipt Digitizer

Milestone 3
Presentation by
Team A

March 3 2026



The Challenge

- Manual data extraction is time-consuming
 - High probability of human errors
 - Unstructured invoice formats vary widely
 - Difficult to store and track physical documents
 - No centralized monitoring system
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Our Goal

To build an intelligent system that:

- Accepts Receipts/Invoices (JPG, PNG, PDF)
- Extracts text using OCR
- Uses LLM for intelligent data understanding
- Stores structured data in database
- Provides a dashboard for monitoring and tracking

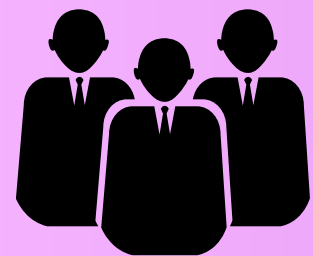


End-to-End Workflow



4.

Dashboard displays results



3.

Data stored in database



1.

User uploads document



2.

- OCR extracts raw text
- LLM processes & structures data

Technologies used

Python

pytesseract (OCR Engine)

Pillow (PIL)

pdf2image

Regex

LLM (Gemini / other)

Flask / Django (Backend)

SQL Database

Frontend Dashboard

OCR Engine

- Multi-format support (.jpg, .png, .pdf)
- Automatic file type detection
- Multi-page PDF support
- Raw text extraction
- Document classification (Invoice / Receipt)
- Line item extraction
- JSON output generation



Data Extraction Capabilities

Invoice Data Extracted

- Invoice Number
- Vendor & Customer Details
- GSTIN
- CGST / SGST / IGST
- Due Date
- Total Amount
- Balance Due

Receipt Data Extracted

- Store Name
- Transaction ID
- Date & Time
- Items Purchased
- Payment Method
- Total Amount



User Authentication System

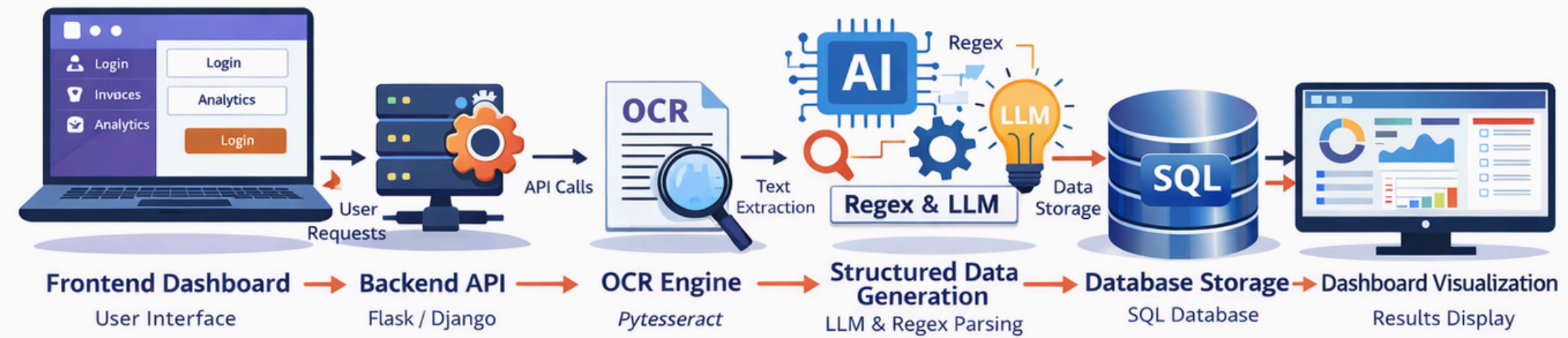
Secure Access Layer

- User Registration
- Login Page
- Session Handling
- User-specific invoice storage
- Secure database mapping

Each user sees only their own
processed documents.



System Architecture



Dashboard Integration

We upgraded from a standalone model to a full application:

- User authentication system
- Web-based file upload
- Database storage
- Invoice history tracking
- Real-time dashboard updates

Dashboard Features

Interactive Dashboard Includes:

- Total invoices processed
- Total receipts processed
- Recently uploaded documents
- Search functionality
- View structured JSON data
- Download extracted data

Invoice History Management

- All processed invoices stored in database
- Filter by:
- Date
- Invoice Number
- Vendor
- View extraction results
- Maintain audit trail

Business Impact

Benefits

- Saves time
- Reduces manual errors
- Centralized invoice storage
- Easy search & retrieval
- Scalable solution
- ERP integration ready

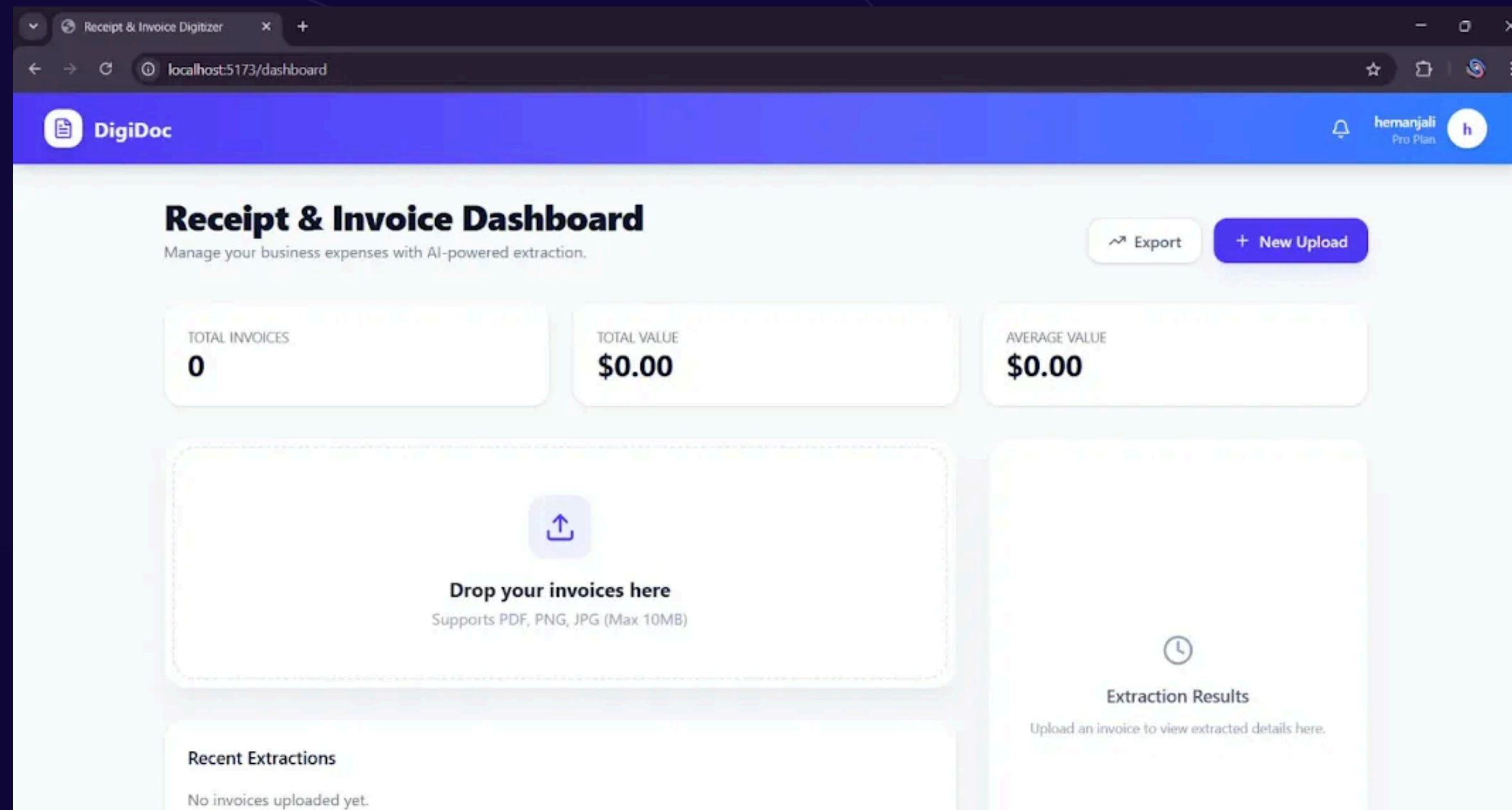
Future Improvements

Planned Enhancements

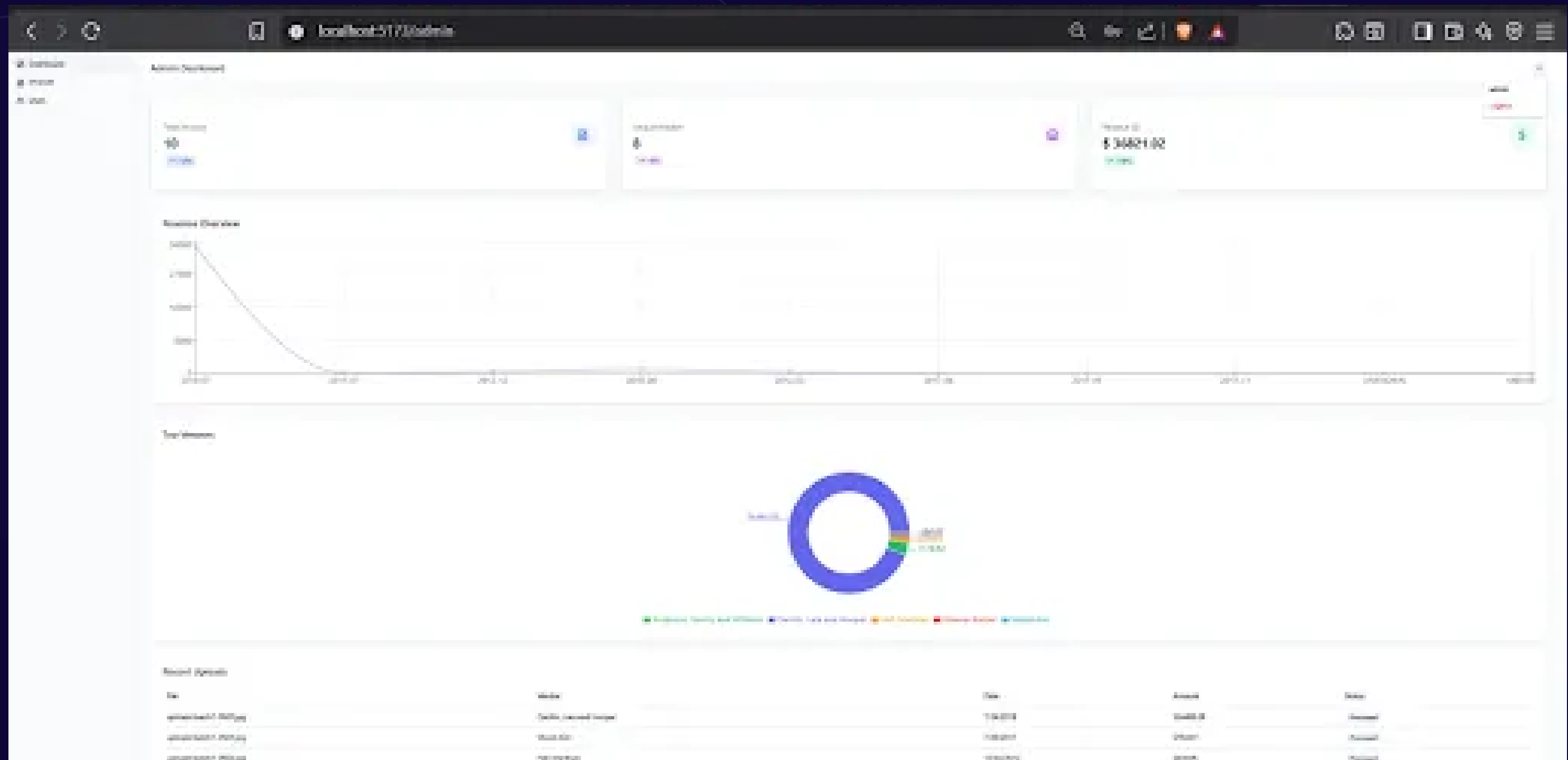
- AI-based layout detection
- Advanced table detection
- Handwritten text recognition
- Cloud deployment (AWS/Azure)
- REST API integration
- Excel export
- Email automation
- Advanced analytics



Current Dashboard page: User



Admin page



Conclusion

Our project successfully transforms manual invoice processing into an intelligent automated system.

By integrating:

- ♦ OCR (pytesseract) for text extraction
- ♦ LLM-based processing for intelligent data understanding
- ♦ Regex validation for structured accuracy
- ♦ Database storage for organized record management
- ♦ Interactive dashboard for monitoring & tracking



Thank You